Irvin Lim

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About Me

Distributed systems engineer focused on Kubernetes and Linux, specializing in workload scheduling and resource isolation.

Currently Tech Lead (Staff-equivalent) of the **Batch Colocation Platform** at Shopee, which runs batch workloads alongside production services on opportunistic idle capacity. Today the platform runs hundreds of thousands of batch workloads on 1M+ reclaimed CPU cores and PBs of memory across more than 40,000 nodes fleet-wide.

Professional Experience

Shopee — Singapore

Senior Expert Engineer, Unified Scheduling (Tech Lead) *Jul 2025 – Present*

- As technical lead of the **Batch Colocation Platform**, delivered tens of millions of dollars in estimated annual savings by scaling resource coverage and onboarding new tenants across six major customers within Big Data, Content and Marketplace businesses. Led cross-team alignment, investigation efforts and architectural changes across team boundaries, driving reclaimed CPU up 4x and memory 6x since 2024 with zero major incidents across P0 campaigns.
- Owned a cross-org **Reverse Colocation** program across seven teams, giving bursty Marketplace services hundreds of thousands of extra CPU cores during sales campaigns by preempting lower-priority Big Data workloads. Scaled service coverage from 10% in a P2 campaign to 80% in P0 campaigns within half a year, letting the business handle last-minute increases in campaign projections without needing to purchase thousands of additional machines.
- Led the architecture and adoption of **Global Scheduler**, a cross-cluster scheduler that manages 100,000+ pods across 50+ services on 40+ clusters. Designed and delivered fixes for the head-of-line blocking and flip-flop loops that blocked adoption, enabling cross-cluster autoscaling to support ad-hoc transcoding demand while cutting resource waste.
- Established the team's quarterly and yearly roadmaps and mentored 10+ teammates through weekly 1:1s, design reviews and technical sharing sessions. Coached and provided growth plans for mentees to own designs and implementations end to end, and grew four team members from individual contributors into component owners.

Expert Engineer, Unified Scheduling (Tech Lead) *Jan 2023 – Jun 2025*

- Led the **Livestream Colocation** project, moving *Shopee Live*'s transcoding off costly public cloud onto internal compute tiered by each streamer's stability needs. Designed and validated pressure-based eviction and rescheduling strategies that made opportunistic capacity safe for live streams, and rolled out aggressive overcommit and colocation, scaling to half a million CPUs and growing transcoding capacity by 5x within a year.
- Drove the isolation evaluation for **Dedicated Pool Colocation** with latency-sensitive *Search, Recommendation & Ads* pools, proposing SMT-aware isolation over static core reservation, and led the stress testing and phased rollout that raised CPU utilization from 15% to 60% with <1 ms added P99 latency.
- Designed and built **Eunomia Meta**'s multi-cluster gateway that provides service discovery, leader-aware routing and load balancing, and serves as a single communication path for the platform's node agents, schedulers and controllers, enabling cross-cluster scheduling and management across hundreds of clusters.
- Led the migration of Helm releases on 100% of clusters to a single declarative, ArgoCD-backed workflow within a quarter, eliminating version and configuration drift and the misconfiguration risk it caused.

Senior Engineer, FinOps & Orchestration *Jan 2022 – Dec 2022*

- Designed the architecture of **Eunomia Agent**, a node agent governing cgroups and cpusets across tens of thousands of nodes today. Built a dual-scheduler model that lets existing batch workloads adopt colocation with minimal changes, and unblocked colocation rollout on clusters by unifying two legacy cgroup agents into a single architecture.
- Drove the initial adoption of **YARN Colocation**, running low-priority Apache Spark and Flink jobs alongside production traffic with CPU, memory, I/O and memory-bandwidth isolation (cgroups, resctrl) that raised peak CPU utilization fourfold with <5% P99 impact, providing estimated cost savings of over a million dollars in its first year.

Software Engineer, Linux Elastic Application Platform *May 2020 – Dec 2021*

- Designed and built **Runonce**, a Kubernetes-native, exactly-once cron and ad-hoc job scheduler that powers millions of job executions per day in production, replacing a legacy scheduler prone to crashing every 2–4 weeks under load.
- Migrated tens of thousands of jobs across the entire company to the new scheduler within a single quarter, while simultaneously supporting 3x year-on-year growth in job volume, and cutting tail task launch latency from 2 minutes to 10 seconds via jittered scheduling techniques.

Internships

Facebook — Menlo Park, CA, USA

Production Engineer Intern, Data Infrastructure *May 2019 – Aug 2019*

- Developed a web service for managing hundreds of production PrestoDB clusters, which replaced the old Presto dashboard with 15x faster load time.

Shopee — Singapore

Software Engineering Intern, TechOps *May 2018 – Oct 2018*

- Designed and developed the first version of **Bromo**, a Mesos-based container orchestrator that evolved onto Kubernetes and now runs millions of pods company-wide.

Conference Talks

Squeeze Your K8s: How We Adopt Time-Series Forecasting Models in FinOps Practices
KubeCon + CloudNativeCon Europe 2024, Paris, France

Colocating Hadoop YARN with Kubernetes to Save Massive Costs on Big Data
KubeCon + CloudNativeCon Europe 2023, Amsterdam, Netherlands

Education

Brown University — Providence, RI, USA

Master of Science, Computer Science *Jan 2019 – May 2020*

GPA: 4.0 / 4.0

National University of Singapore — Singapore

Bachelor of Computing (Honours), Computer Science *Aug 2015 – Dec 2018*

CAP: 4.58 / 5.00, Dean's List

Skills

- **Languages**: Go (primary), C, Python, Java, JavaScript
- **Linux Kernel**: Performance profiling (eBPF, perf, PSI, RDT), runtime isolation (cgroups, resctrl, NUMA/SMT), Linux task scheduling, Linux memory management
- **Resource Management**: FinOps, workload colocation, multi-tenancy and QoS, overcommit, eviction and preemption, capacity planning, time-series forecasting
- **Kubernetes**: Cluster management, application delivery (Helm, ArgoCD), customized controller and scheduler plugin development, node & container runtime (Docker, containerd, NRI plugins)
- **Distributed Systems**: Multi-level and job scheduling, cluster federation, high availability and leader election, service discovery and load balancing
- **Observability & Data**: Prometheus, VictoriaMetrics, Grafana, MySQL, PostgreSQL, Redis
- **Language Fluency**: English (native), Mandarin Chinese (native)